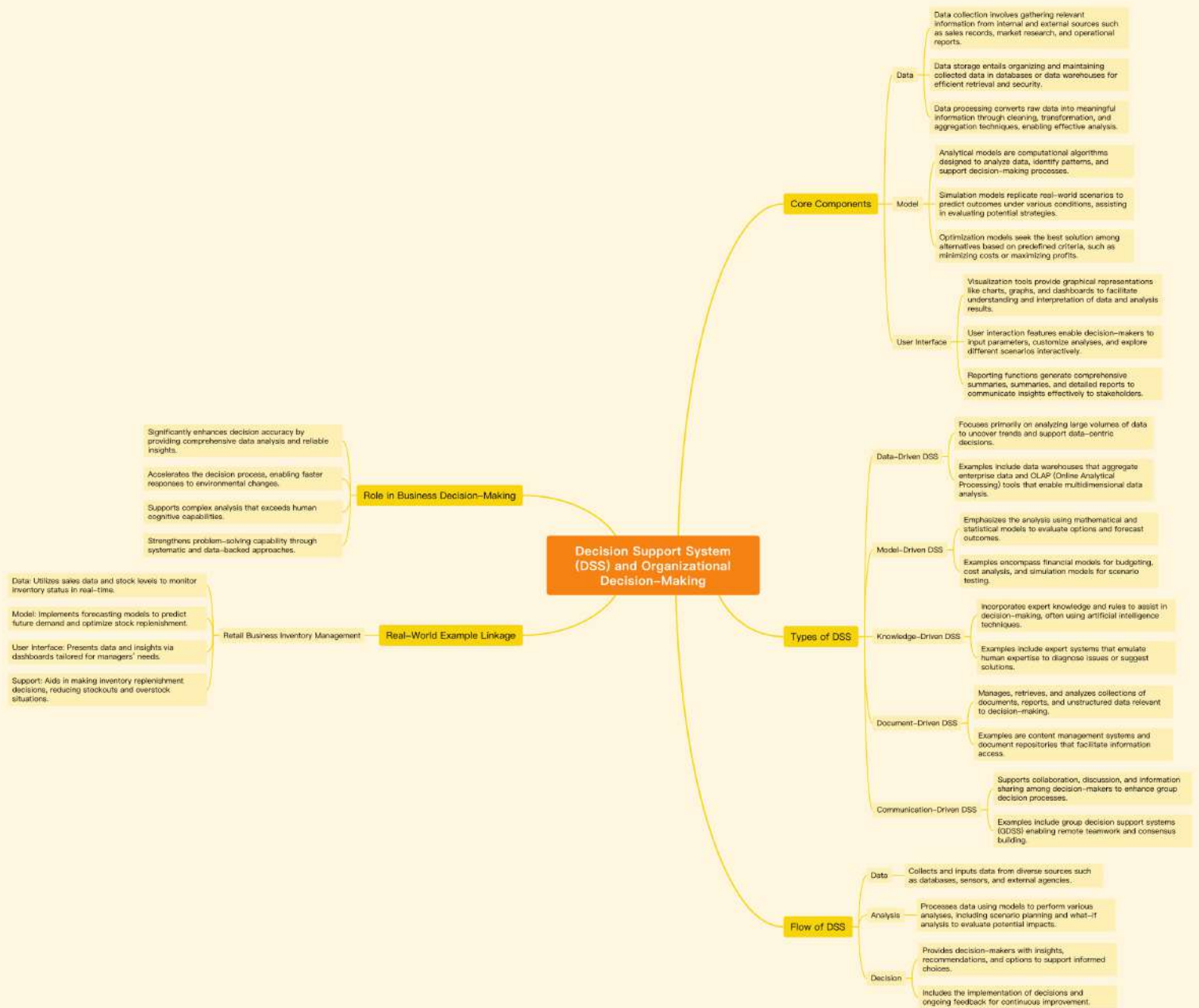
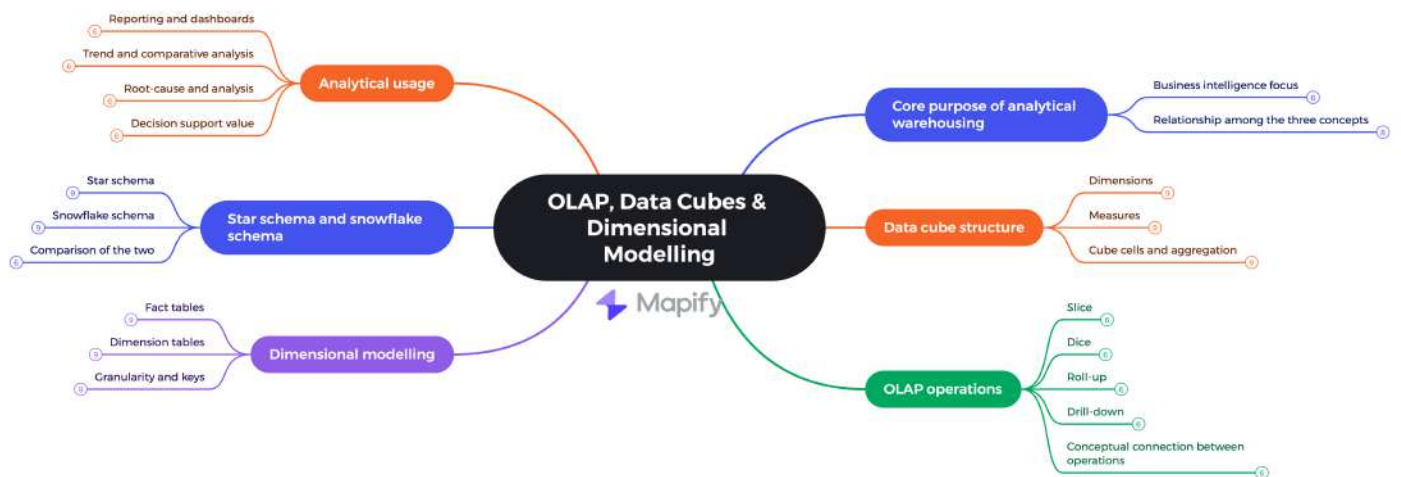
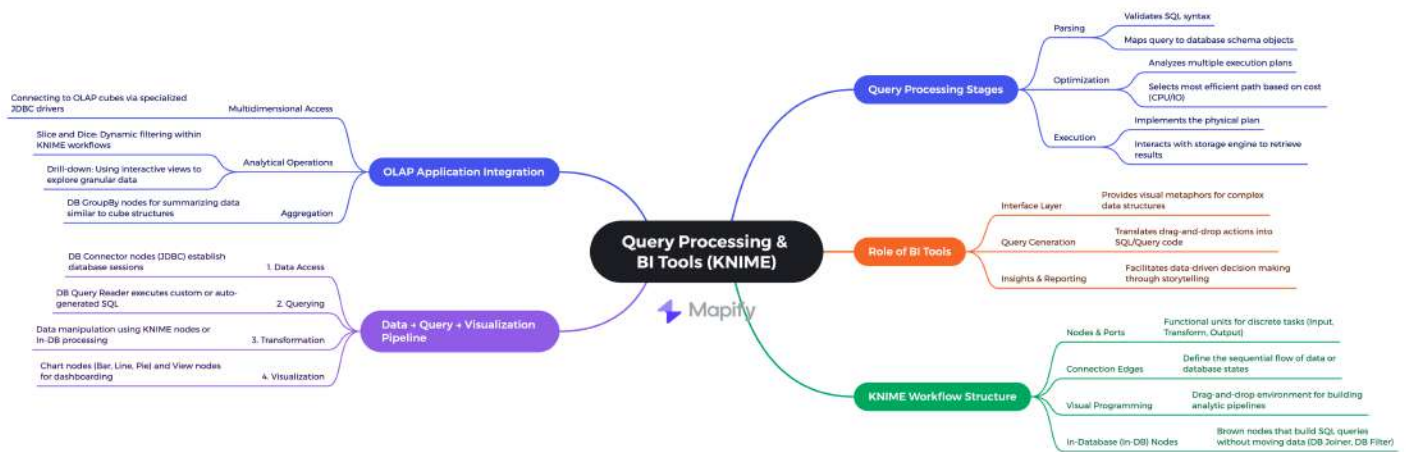








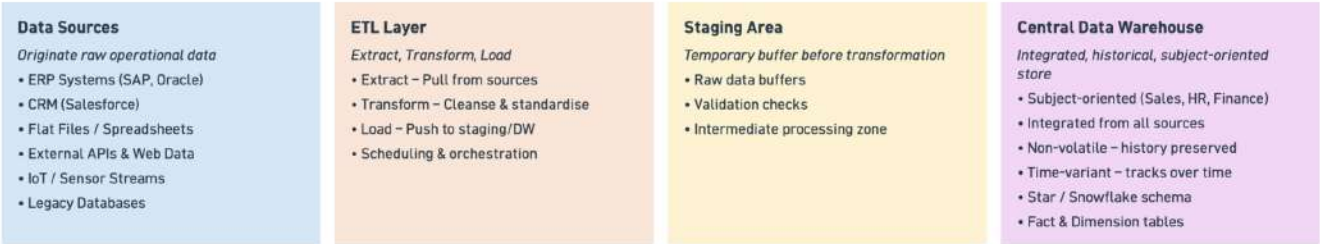
Architecture of a Data Warehouse

Layers · Data Flow · Component Roles

Data Flow Overview (Source → End User)



Layer Details — Sources → Warehouse



Metadata Layer — The Knowledge Backbone



Data Marts & Front-End Tools



Role of Each Layer — Summary

Layer	Primary Role	Data State
Data Sources	Originates raw, live transactional data from operational systems	Raw / Live
ETL Layer	Extracts, transforms and loads data into the warehouse	In-flight
Staging Area	Temporary buffer for raw data before transformation	Unvalidated
Central Data Warehouse	Centralised, integrated, historical subject-oriented store	Integrated
Metadata Layer	Describes structure, origin and meaning of all warehouse data	Descriptive
Data Marts	Subject-specific subsets for departmental analytical use	Aggregated
Front-End Tools	Business users access data via reports, OLAP and dashboards	Presented

Operational Systems vs Decision Support Systems

OLTP ↔ OLAP / DSS

Definitions

OLTP – Online Transaction Processing
Manages day-to-day operational transactions in real time. Ensures data integrity via ACID properties.

DSS / OLAP – Decision Support System
Supports complex analytical queries on large historical data.

Key Differences

Dimension	OLTP (Operational)	OLAP / DSS (Analytical)
Data Type	Current, real-time, detailed records	Historical, aggregated, summarised data
Purpose	Record & process daily transactions	Analyse trends & support decisions
Users	Clerks, front-line staff, applications	Analysts, managers, executives
Processing	Simple INSERT/UPDATE, high concurrency	Complex queries, joins, aggregations
Response	Milliseconds (real-time)	Seconds-minutes (batch/warehouse)
DB Design	Normalised (3NF) – avoids redundancy	Denormalised / Star / Snowflake schema

Similarities

- Both store and retrieve structured data
- Both rely on relational/SQL-based engines
- OLTP feeds raw data → ETL → OLAP/Data Warehouse
- Both require data quality & security

Real-Time vs Historical Data

Real-Time Data (OLTP) • Captures transactions as they happen • Data is current and volatile • Example: a bank balance updates instantly after every transfer	→ ETL →	Historical Data (OLAP) • Aggregated over months/years • Data is stable, read-heavy • Example: quarterly revenue trends in a BI dashboard
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How They Interact – Data Flow



Example Systems

OLTP Examples

- Core Banking**
Account credits, debits, transfers
- POS / Retail**
Sales at checkout, inventory updates
- Hospital EMR**
Patient admissions, prescriptions

DSS / OLAP Examples

- Sales Analytics**
Revenue trends, region performance
- Business Intelligence**
Dashboards, KPIs, forecasting
- Risk & Fraud Analysis**
Pattern detection, anomaly scoring